

## PARABURDOO, Australia

WMO: 943160

Lat: 23.1725S

Lon: 117.7494E

Elev: 1392

StdP: 13.97

Time Zone: 8.00 (E08)

Period: 96-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 45.8       | 48.2 | 9.3                           | 9.3 | 78.0 | 12.9 | 11.1 | 76.8 | 19.5                  | 65.7 | 18.2 | 66.0 | 6.2                   | 60   | 0.455 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |       |      |       |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|-------|------|-------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%    |      | 2%    |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB    | MCWB | DB    | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)   | (f)  | (g)   | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 22.9                   | 109.2           | 66.2 | 107.1 | 65.8 | 105.0 | 65.6 | 76.5                | 82.8 | 75.5 | 83.6 | 74.5 | 84.4 | 11.1                 | 260  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 75.5                            | 140.6 | 78.0 | 73.8 | 132.6 | 78.3 | 72.0 | 124.8 | 79.1 | 41.3          | 82.2 | 40.1 | 82.7 | 39.2 | 84.3 | 82.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |     |
| (4) 20.9          | 18.6 | 16.7 | DB                         | 40.6 | 112.4              | 2.8 | 1.8                                                | 38.5 | 113.8      | 36.8 | 114.8      | 35.2 | 115.9      | 33.2 | 117.2 | (4) |
| (5)               |      |      | WB                         | 33.6 | 78.3               | 2.0 | 2.0                                                | 32.1 | 79.7       | 30.9 | 80.9       | 29.8 | 82.0       | 28.3 | 83.5  | (5) |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 78.2   | 90.9  | 88.3  | 85.6  | 79.7  | 71.1  | 64.5  | 63.3  | 66.5  | 72.3  | 80.4  | 85.9  | 90.2  |
|                                                                               | DBStd       | 10.96  | 5.41  | 5.20  | 5.06  | 4.72  | 5.01  | 4.57  | 4.25  | 4.32  | 4.59  | 4.85  | 4.78  | 4.58  |
|                                                                               | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                               | HDD65       | 195    | 0     | 0     | 0     | 0     | 8     | 65    | 85    | 33    | 4     | 0     | 0     | 0     |
|                                                                               | CDD50       | 10280  | 1268  | 1073  | 1104  | 892   | 653   | 434   | 411   | 510   | 669   | 944   | 1076  | 1246  |
|                                                                               | CDD65       | 5000   | 803   | 653   | 639   | 442   | 196   | 49    | 31    | 78    | 223   | 479   | 626   | 781   |
|                                                                               | CDH74       | 71953  | 12564 | 9622  | 8875  | 5238  | 1922  | 410   | 431   | 1153  | 2996  | 6813  | 9393  | 12536 |
|                                                                               | CDH80       | 43353  | 8446  | 6063  | 5261  | 2684  | 683   | 52    | 58    | 319   | 1368  | 3978  | 5964  | 8477  |
| Wind                                                                          | WSAvg       | 8.5    | 9.1   | 8.8   | 8.6   | 7.8   | 8.0   | 8.1   | 7.9   | 8.2   | 8.4   | 8.8   | 9.3   | 9.2   |
| Precipitation                                                                 | PrecAvg     | 10.5   | 2.0   | 2.4   | 1.3   | 0.9   | 1.0   | 0.9   | 0.5   | 0.3   | 0.1   | 0.1   | 0.3   | 1.0   |
|                                                                               | PrecMax     | 18.0   | 7.9   | 11.9  | 5.3   | 6.0   | 4.7   | 4.2   | 3.5   | 1.9   | 1.8   | 0.8   | 1.2   | 7.1   |
|                                                                               | PrecMin     | 2.2    | 0.0   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   |
|                                                                               | PrecStd     | 4.0    | 2.1   | 2.5   | 1.5   | 1.3   | 1.3   | 1.1   | 0.7   | 0.5   | 0.3   | 0.2   | 0.3   | 1.4   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 111.7 | 110.8 | 106.1 | 101.3 | 93.1  | 84.5  | 84.6  | 89.7  | 96.0  | 103.9 | 106.6 | 111.5 |
|                                                                               |             | MCWB   | 66.2  | 65.6  | 66.7  | 64.4  | 61.3  | 59.6  | 57.9  | 58.5  | 59.7  | 62.6  | 63.9  | 65.9  |
|                                                                               | 2%          | DB     | 109.6 | 107.7 | 103.1 | 97.2  | 90.0  | 81.3  | 81.5  | 86.5  | 93.5  | 100.7 | 104.3 | 108.5 |
|                                                                               |             | MCWB   | 66.4  | 66.3  | 66.5  | 63.9  | 60.4  | 58.2  | 56.9  | 56.9  | 58.6  | 61.7  | 63.3  | 65.6  |
|                                                                               | 5%          | DB     | 107.4 | 104.9 | 100.8 | 94.7  | 87.0  | 78.7  | 78.7  | 83.7  | 91.1  | 98.3  | 102.4 | 106.2 |
|                                                                               |             | MCWB   | 66.2  | 66.5  | 66.5  | 63.0  | 59.9  | 57.0  | 55.4  | 56.1  | 58.1  | 61.1  | 62.9  | 65.2  |
|                                                                               | 10%         | DB     | 104.8 | 101.4 | 98.1  | 92.2  | 84.0  | 76.0  | 76.0  | 80.8  | 88.1  | 95.8  | 100.3 | 103.7 |
|                                                                               |             | MCWB   | 66.8  | 67.1  | 66.9  | 62.8  | 59.3  | 55.6  | 54.7  | 55.3  | 57.7  | 60.7  | 62.8  | 64.9  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 78.6  | 78.7  | 76.9  | 73.2  | 70.7  | 67.8  | 64.9  | 64.2  | 64.7  | 68.2  | 70.1  | 75.4  |
|                                                                               |             | MCDB   | 80.9  | 81.0  | 80.2  | 79.2  | 75.3  | 70.6  | 70.8  | 71.8  | 79.7  | 85.0  | 83.6  | 81.9  |
|                                                                               | 2%          | WB     | 76.4  | 76.8  | 75.0  | 71.2  | 66.0  | 64.5  | 62.1  | 61.3  | 62.1  | 66.1  | 68.0  | 73.5  |
|                                                                               |             | MCDB   | 83.1  | 83.2  | 82.0  | 80.1  | 73.7  | 72.2  | 70.7  | 75.7  | 83.8  | 85.3  | 89.4  | 84.8  |
|                                                                               | 5%          | WB     | 75.3  | 75.7  | 73.7  | 69.8  | 64.1  | 62.3  | 59.7  | 59.2  | 60.3  | 64.5  | 66.4  | 72.1  |
|                                                                               |             | MCDB   | 85.1  | 83.4  | 83.3  | 80.4  | 76.5  | 71.6  | 70.8  | 77.8  | 84.3  | 87.3  | 91.0  | 86.4  |
|                                                                               | 10%         | WB     | 74.3  | 74.7  | 72.5  | 68.2  | 62.4  | 60.1  | 57.5  | 57.3  | 59.0  | 62.9  | 65.1  | 70.6  |
|                                                                               |             | MCDB   | 86.0  | 83.9  | 83.8  | 80.7  | 77.6  | 70.5  | 70.4  | 76.9  | 84.5  | 90.0  | 91.2  | 88.2  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 22.9  | 21.6  | 21.4  | 22.6  | 23.3  | 22.4  | 24.5  | 27.2  | 29.2  | 28.5  | 26.1  | 24.7  |
|                                                                               |             | MCDBR  | 26.9  | 27.3  | 25.3  | 25.8  | 26.7  | 25.3  | 29.1  | 31.9  | 33.8  | 31.7  | 28.5  | 27.6  |
|                                                                               |             | MCWBR  | 7.4   | 7.0   | 6.6   | 8.2   | 9.8   | 10.7  | 12.5  | 13.4  | 12.9  | 11.2  | 8.9   | 8.5   |
|                                                                               | 5% WB       | MCDBR  | 20.4  | 18.9  | 18.6  | 18.6  | 19.4  | 17.3  | 21.0  | 25.9  | 29.2  | 25.8  | 24.9  | 22.1  |
|                                                                               |             | MCWBR  | 5.5   | 5.1   | 6.0   | 6.1   | 8.5   | 8.8   | 10.6  | 12.2  | 11.9  | 8.5   | 7.6   | 6.1   |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.367  | 0.366 | 0.356 | 0.340 | 0.315 | 0.301 | 0.303 | 0.308 | 0.324 | 0.345 | 0.344 | 0.351 |       |
|                                                                               | taud        | 2.554  | 2.572 | 2.574 | 2.569 | 2.568 | 2.593 | 2.561 | 2.512 | 2.467 | 2.445 | 2.482 | 2.545 |       |
|                                                                               | Ebn at Noon | 310    | 306   | 300   | 292   | 287   | 286   | 289   | 298   | 305   | 309   | 315   | 315   |       |
|                                                                               | Edh at Noon | 35     | 34    | 32    | 30    | 27    | 25    | 27    | 31    | 35    | 37    | 37    | 35    |       |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 2373   | 2244  | 1991  | 1719  | 1469  | 1306  | 1406  | 1782  | 2148  | 2382  | 2554  | 2556  |       |
|                                                                               | RadStd      | 149    | 141   | 139   | 89    | 108   | 100   | 116   | 66    | 72    | 113   | 112   | 153   |       |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only           | +1.76 | +1.66   | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page